Detector Resonant Frequencies Analysis in PICO-40L

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Abstract

The PICO-40L experiment is a bubble chamber detector in search of Weakly Interacting Massive Particles (WIMPs), potential candidates for dark matter. A key feature of the PICO-40L design is its acoustic discrimination system, which differentiates alpha particle radiation events from potential WIMP candidates. This system utilizes piezoelectric transducers to record acoustic signals from bubble formations. However, the noise characteristics of the detector, including the resonant frequencies of the detector's stainless steel bellows, have not yet been studied. We analyze the piezoelectric sensor data to identify and characterize these resonant frequencies. Using the bellows movement during compression as an impulse, we compute power spectral densities of the acoustic signals using Welch's method. Our analysis reveals excited resonances that are coupled with the bellows' position, suggesting that these resonances originate from the bellows. By understanding our detector noise, we can better identify and characterize our signals, as well as improve our understanding of the acoustics system.

PHYS 499 - Undergraduate Research Project
University of Alberta
April 2024

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1 Introduction and Background

1.1 The Dark Matter Problem

The existence of dark matter remains one of the biggest open problems in modern physics. Although evidence for its existence is abundant—seen in phenomena like the motion of galaxies, the structure of the cosmic microwave background, and the formation of the Bullet Cluster—its exact nature remains unknown [1] [2]. Weakly Interacting Massive Particles (WIMPs) are among the most promising dark matter candidates. WIMPS offer a theoretical framework that aligns with both astrophysical observations and particle physics models. However, despite extensive international efforts, WIMPs have not yet been detected directly, leaving the dark matter puzzle unsolved. The PICO-40L experiment is an ongoing effort at SNOLAB to resolve this mystery.

1.2 PICO-40L Bubble Chamber

Invented in 1952 by Donald Glaser [3] to study cosmic ray showers, bubble chambers have proven to be powerful tools for detecting rare particle interactions. The fundamental principle of the bubble chamber is that it contains a superheated fluid within a sealed environment. When a particle interacts with this fluid and imparts energy, bubbles form. This phenomenon occurs because the fluid and the detector lacks nucleation sites for bubble formation, so the deposited energy acts as a nucleation trigger.

The PICO-40L experiment employs a right-side-up bubble chamber design [1]. Its inner vessel (IV), constructed from quartz, contains the target material, C_3F_8, and is sealed to stainless steel bellows. This IV is housed within a pressure vessel (PV) filled with a fluid that transmits pressure to the target fluid through the bellows. Under current operating conditions, the detector maintains the center of the target fluid at a pressure of (27.5 ± 0.1) psia [1]. This pressure was carefully selected to optimize several factors, including nucleation threshold energy to minimize background events while maximizing detection of potential WIMP interactions, acoustic performance, and other considerations [1]. The operating temperature is maintained at (13 ± 1)°C, ensuring that the effective "boiling point", or the energy required for nucleation, is (2.855 ± 1) keV [1].

Crucially, because the bellows are in direct contact with the target fluid, their resonant frequencies are directly coupled to the active material. As a result, any acoustic systems employed in the detector will inherently reflect the resonances of the bellows.

1.3 Acoustic System in PICO-40L

Acoustic systems play a critical role in PICO-40L. One of the key advantages of the bubble chamber design is its ability to acoustically discriminate alpha-particle-induced events from potential WIMP candidate events [4]. A large source of background events in the detector is from alpha particles produced by the decay of environmental ^{222}Rn. The nucleation of bubbles from alpha interactions is visually indistinguishable from those caused by WIMPs.

To address this, PICO-40L is equipped with twelve piezoelectric transducers (piezos) that capture acoustic signals for each event at a sampling frequency of 1 MHz over a duration of 1.5 seconds, which helps define an acoustic parameter. Alpha-particle interactions typically produce higher acoustic power, allowing them to be rejected during data analysis. Among these piezos, eight are 'normal', while the others are specialized variants: one is optimized for higher-frequency response, and the other omits the insulating epoxy present in the normal piezo. These piezos are strategically placed at different locations and angles on the detector to ensure comprehensive coverage and avoid frequency nodes.

Despite the importance of the acoustic system in discriminating events, the noise characteristics of the detector have yet to be studied. This is particularly important because the bellows, which are large metal

springs in direct contact with the target material, can introduce acoustic noise through their movement. Thus motivating our study of the resonant frequencies of the bellows.

2 Methodolgy

The primary challenge in our analysis is determining the resonant frequencies of the detector remotely. The simplest method to identify resonant frequencies is through an impulse, which acts as a broadband excitation. Our approach is to analyze the piezoelectric traces corresponding to the movement of the bellows after a bubble event (see Fig. 1). Following a bubble event, the bellows compress the target fluid to return it to the stable state. We treat this bellows movement as an impulse, examining the frequency spectrum after the compression compared to the pre-event state. The difference in power spectral densities (PSDs) reveals which frequencies are excited.

If we demonstrate that these frequencies are coupled to the bellows' position and share similar quality factors—indicating they originate from the same component of the detector—we can confidently attribute them to the bellows.

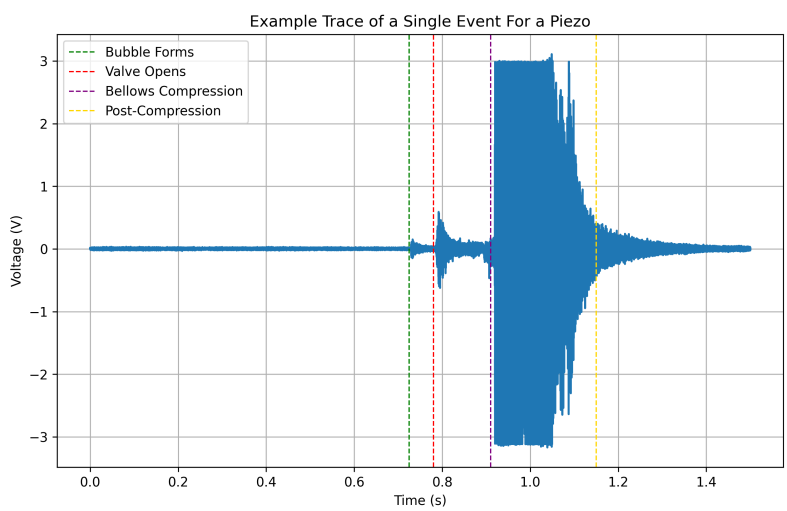

Figure 1: An acoustic signal from a piezo trace. The regions of interest are indicated on the trace. We primarily inspect the frequencies pre-bubble formation and post-bellows compression to find the excited frequencies.

For a comparative analysis, we focus on piezo 0 and piezo 19, enabling a direct comparison between two distinct locations. In particular, for our analysis, piezo 0 and piezo 19 are 60° apart, and are separated by two layers of plastic rings [1]. Since each piezo operates at a sampling frequency of 1 MHz, the Nyquist frequency (maximum detectable frequency) is 500 kHz. We analyze the voltage traces from 30 events, resulting in 60 traces—one from each piezo per event. To estimate the power spectral density of the stationary signals, we employ Welch's method, which is described as follows [5]:

1. **Divide the Signal into Overlapping Segments:** Let $x(t)$ be the input signal, and N the total length of the signal. Welch's method divides the signal into M overlapping segments, each with length

L, where we use $L = 1024$ in our analysis. The segments overlap by $\frac{L}{2}$ points. Mathematically, we denote the segment signals as:

$$x_1(t), x_2(t), \ldots, x_M(t)$$

2. **Apply a Window Function:** Each segment $x_m(t)$ is windowed to reduce spectral leakage. A window function $w(t)$ is applied to each segment:

$$x'_m(t) = x_m(t) \cdot w(t)$$

In our case, we use the Hann window function [6] [7].

3. **Compute the Periodogram for Each Segment:** For each windowed segment, the periodogram is computed as the squared magnitude of the discrete Fourier transform:

$$P_m(f) = \left|\mathcal{F}\left\{x'_m(t)\right\}\right|^2$$

where $\mathcal{F}\{\cdot\}$ denotes the discrete Fourier transform.

4. **Average the Periodograms:** The final PSD estimate is obtained by averaging the periodograms from each segment:

$$\hat{P}(f) = \frac{1}{M} \sum_{m=1}^{M} P_m(f)$$

This averaging reduces the variance of the estimate compared to using a single periodogram of the entire signal.

5. **Normalize the Result:** To account for the windowing, we must normalize our results, scaling it by the sum of the squared window values [8]:

$$\hat{P}(f) = \frac{1}{M} \sum_{m=1}^{M} \frac{P_m(f)}{\sum_t w^2(t)}$$

where $w(t)$ is the window function applied to each segment.

Welch's method is essentially a sliding window average of the discrete fourier transform (DFT) over an interval. Compared to doing a DFT over the whole signal, Welch's method is more sensitive to stationary signals present over the whole interval. This means that we reduce noise in our spectrum, but this is at a cost of spectral resolution.

We plan to estimate the power spectral density (PSD) for multiple intervals within the piezo trace. Using a sliding interval post-compression, we will analyze how the PSD evolves over time. By examining the decay of spectral peaks, we can calculate the decay constant and determine the quality (Q) factor for the identified frequencies. Additionally, we will apply Welch's method to the pre-event data to estimate the background, enabling us to perform background subtraction. Lastly, we use `scipy`'s `find_peaks` method to identify the frequency peaks in the PSDs.

3 Results

After compression, following background subtraction, we observe that the impulse does not excite all frequencies equally. Instead, distinct peak frequencies are evident.

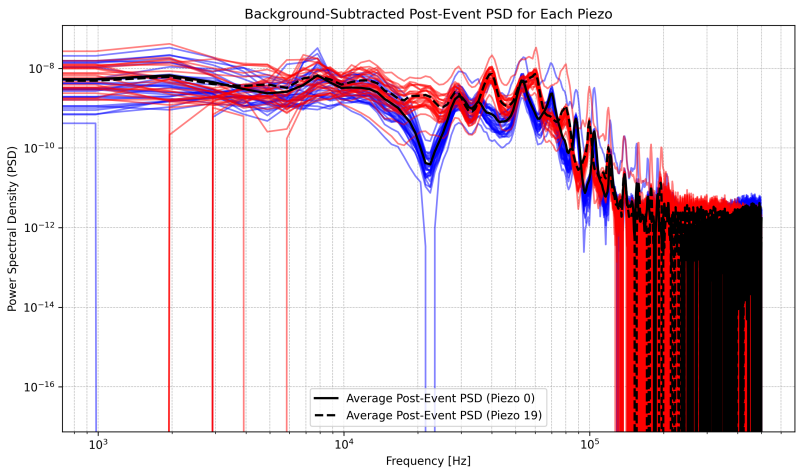

Figure 2: A Welch estimate of the power spectral density for the last 200 ms of a piezo trace, background-subtracted by the Welch estimate of the power spectral density from the first 200 ms of the trace. Each red line represents an event from Piezo 19, and each blue line represents an event from Piezo 0. The black lines indicate the averages.

From Fig. 2, we observe discrepancies in the peak locations. Some peaks are present for Piezo 0 but absent for Piezo 19, and vice versa. To the best of our knowledge, PICO-40L does not have published Bode plots for the frequency response of the piezos, so we cannot definitively attribute these discrepancies to the piezo frequency response or other factors. Notably, there is a significant discrepancy at approximately 20 kHz. Additionally, the excited frequencies in the plot only begin to visibly emerge for frequencies higher than 25 kHz. There is noticeable saturation, with numerous peaks and substantial high-frequency noise at frequencies higher than 100 kHz.

Next, we identify the peak frequencies for each trace in every event. Specifically, we select frequencies f_o that are present in at least 10 other events within 1 kHz of f_o. The tolerance of 1 kHz is motivated by our spectral resolution of 1 kHz, which is constrained by Welch's method.

For each piezo, we tabulate these peaks observed (after compression):

Piezo 0 Frequency Groups (±1 kHz)	**Piezo 19 Frequency Groups (±1 kHz)**
29	28
35	39
45	53
53	60
69	67
88	79
99	88
	91
	100

Table 1: Frequencies Peaks for Piezo 0 and Piezo 19 between 25 kHz and 100 kHz. A peak is only included if there are at least 10 events containing that peak. The uncertainty in the frequency group represents our tolerance for frequencies within this group.

We can visualize the decay of these peak frequencies for each piezo over time by performing a background-subtracted sliding interval Welch PSD. Specifically, we create 100 intervals, each 100 ms in length, starting from 300 ms and extending to 100 ms before the end of the trace. We then observe the PSD value of each frequency peak within these intervals. These results can be seen in Fig. 3.

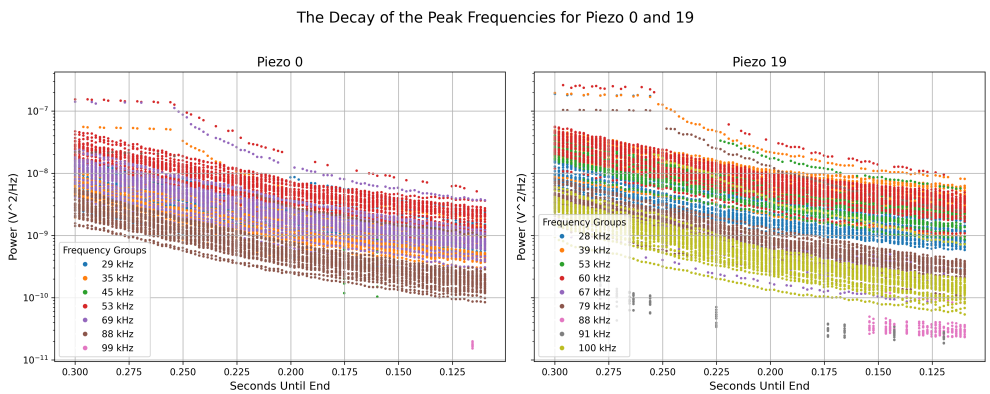

Figure 3: Decay of the peaks over time. For each frequency group, the PSD is tracked from -300 ms to -100 ms, with 100 points sampled until the trace ends. Each color represents a different frequency group.

We can then do an exponential fit on each frequency group decay to obtain a decay constants. This can be seen in Fig. 4 and Fig. 5.

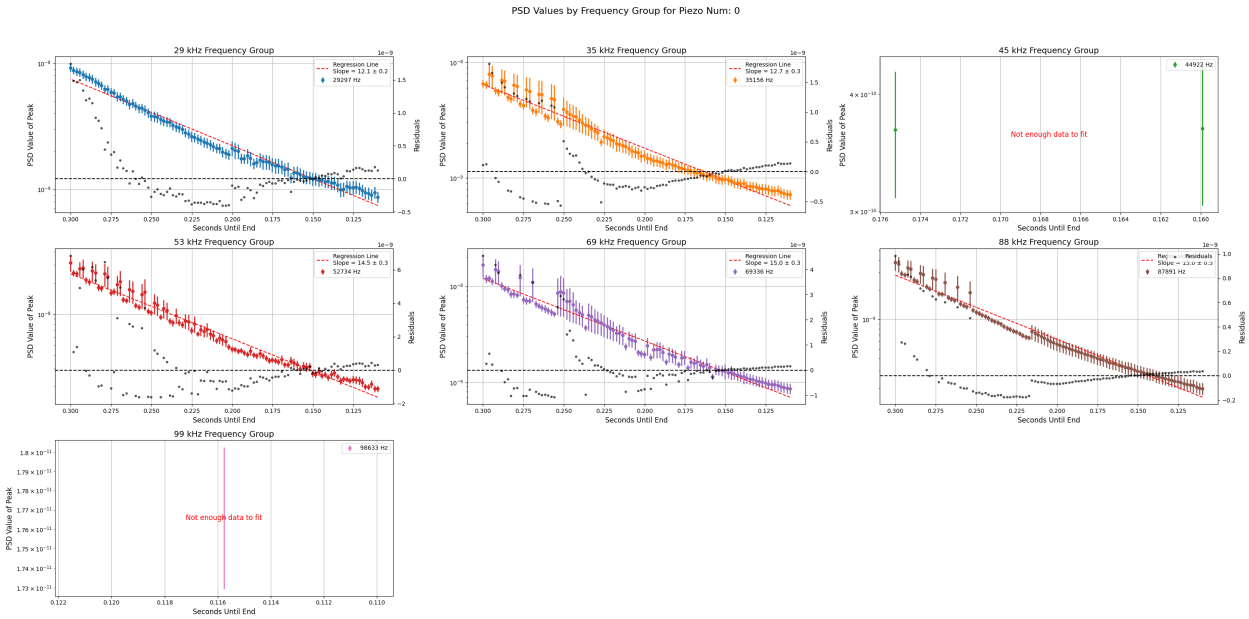

Figure 4: Exponential fits and residuals for Piezo 0. Note that some peaks do not have enough points to justify a fit (we require at least 3 points). The uncertainty in power is due to sample variance.

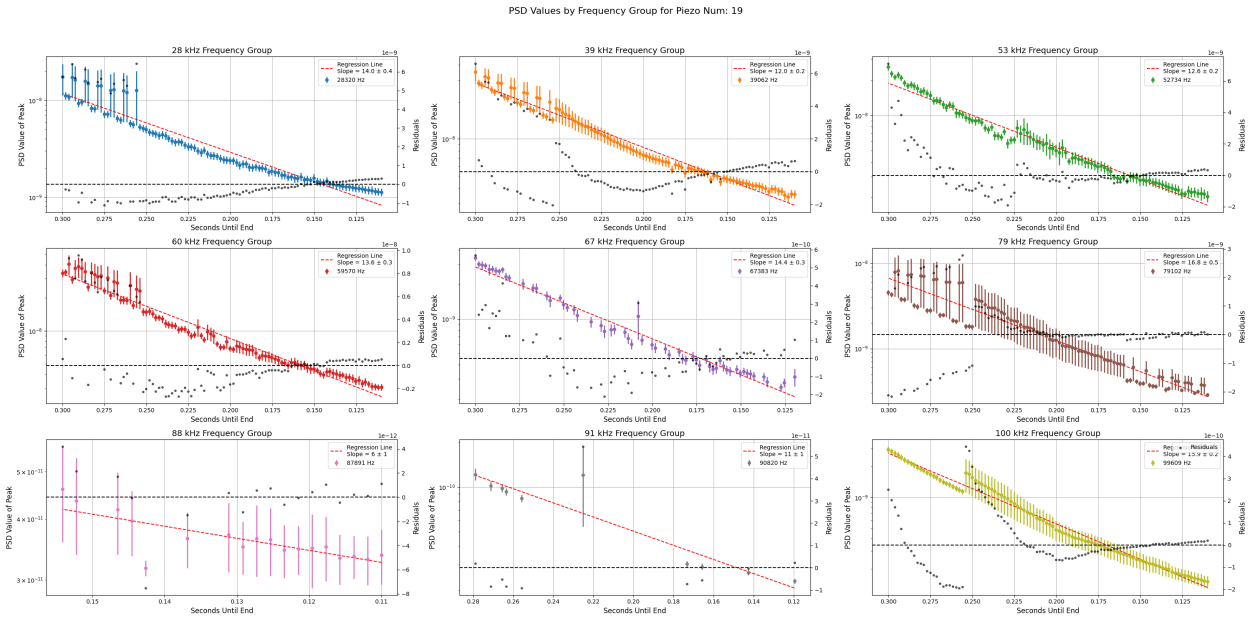

Figure 5: Exponential fits and residuals for Piezo 19. The uncertainty in power is due to sample variance.

We observe a beat frequency, indicating an exchange of energy between two parts of the detector. This beat frequency occurs on the order of 100 Hz, which we cannot spectrally resolve due to our uncertainty. However, we can analyze the overall slopes, our decay constants, to assess the decay constants and Q factors, seen in Fig. 6.

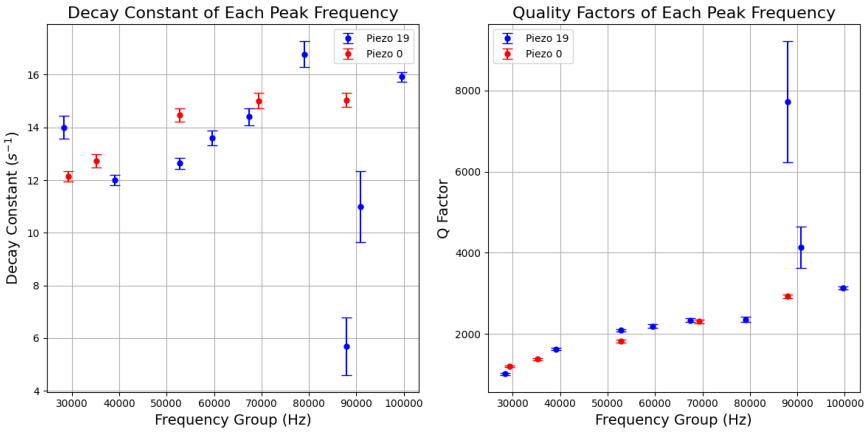

Figure 6: The calculated decay constants and Q factors for the decay of each peak. The uncertainties are derived from the covariance matrix of the slope fit. Note that two of these decay constants exhibit significantly higher variance and are more dissimilar to the others. This is because these peaks are not well-sampled and do not consistently appear in the PSD.

Lastly, we can examine how the power of the peaks is coupled to the average bellows position within the trace. Specifically, we analyze the background-subtracted final 0.2-second interval of the trace.

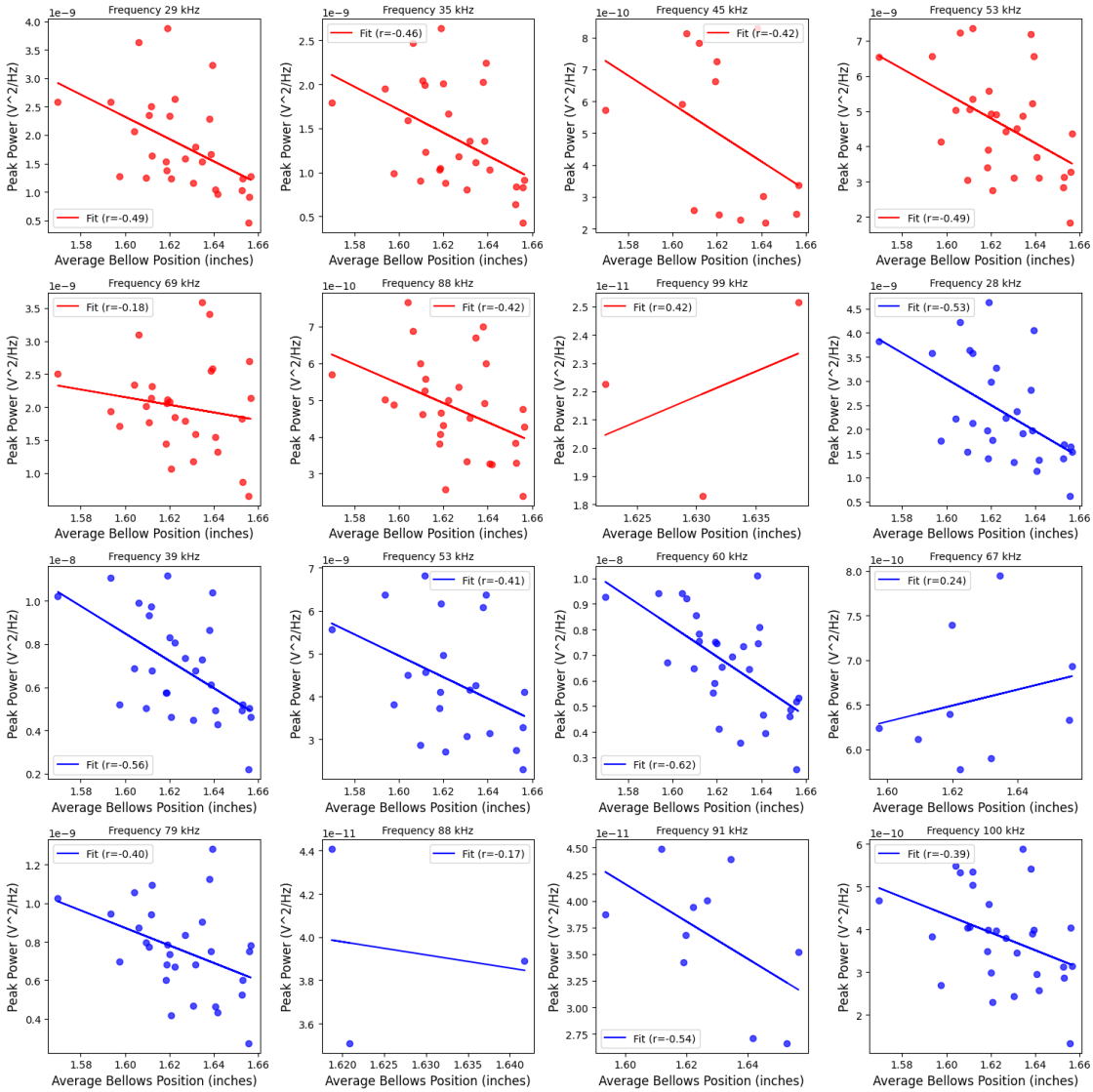

Figure 7: The background-subtracted acoustic power of each peak in the last 200 ms of the trace. Blue represents Piezo 19, and red represents Piezo 0. The Pearson correlation coefficient is indicated for each fit. An outlier event, with power significantly higher than the others (sometimes by an order of magnitude), was excluded from the regression analysis as it was inconsistent with the remaining events. This anomaly may have been caused by lab noise or another external factor. Note: Some frequencies are not well-sampled, but are included for completeness. This is discussed further below.

The data above is summarized in the following table:

Frequency (± 1 kHz)	R Value
Piezo 0	
29	-0.49
35	-0.46
45	-0.42
53	-0.49
69#	-0.18
88	-0.42
99*	0.42
Piezo 19	
28	-0.53
39	-0.56
53	-0.41
60	-0.62
67#	0.24
79	-0.40
88*	-0.17
91	-0.54
100	-0.39

Table 2: Pearson correlation coefficients for each peak and its correlation with the bellows position. Frequencies marked with an asterisk (*) have a weak or no moderate-to-strong negative correlation with the bellows position. These marked frequencies are the poorly sampled results in Figure 7. The frequencies marked by a hashtag (#) are moderately sampled, but still do not show a moderate-to-strong negative correlation with bellows position. This is further discussed below.

4 Discussion

4.1 Discussion of Results

From Fig. 2, we observe that there are peaks, consistent between events, above the background after compression. This indicates that the excitation is not merely broadband. These are excited peaks. However, we also notice that these peaks are not always consistent across the piezos. This is evident in Table 1 listing the peaks between 25 kHz and 100 kHz. The piezos show a different number of total peaks in this range. It is difficult to attribute these discrepancies to the piezo frequency response or other factors, as PICO-40L has not quantified the frequency responses of their piezos.

Despite this, the peaks we identified are often consistent across events for the same piezo, and their decay patterns are relatively consistent as well. We can see in Fig. 4 and Fig. 5 that they steadily decay as expected. However, we do not observe normality of residuals. As a result, linear regression may not fully capture the underlying behavior. A possible explanation for this is most clearly seen in the 79 kHz peak in Piezo 19, shown in Fig. 5. The power of the peak oscillates above and below the linear regression line seven times in the first 50 ms, a frequency on the order of 100 Hz. This indicates that the peak frequency is beating with another frequency. This phenomenon is also observed in other frequency groups with the Welch method, both in Piezo 0 and Piezo 19. Although we lack the spectral resolution to analyze these beats in greater detail, we can confirm their presence. This suggests that some parts of the detector are interacting with each other and exchanging energy continuously.

Nevertheless, we can use the slope of the linear regression as an approximation of the decay constants. Since we are fitting a logarithmic plot, this corresponds to an underlying exponential relation. In principle,

if we showed that the quality factors for each frequency group are similar, this would indicate that they originate from the same component of the detector, where damping would be similar. However, from Fig. 6, we observe that the Q factors are not in good agreement. This result leaves us inconclusive of the origin of the peaks. They may be from different components or different vibrational modes of the same component.

From Fig. 7 and Table 2, we observe that most of the peaks exhibit a moderate to high degree of correlation with the average position of the bellows. Specifically, all peaks except for the ones at (69 ± 1) kHz and (99 ± 1) kHz in Piezo 0, and (67 ± 1) kHz and (88 ± 1) kHz in piezo 19, show a moderate to high negative correlation between their power and the average position of the bellows in the trace. The peaks at (88 ± 1) kHz and (99 ± 1) kHz in Fig. 7 have fewer data points compared to the other peaks, making it difficult to draw any conclusive observations about the correlation. However, both of these peaks have corresponding peaks in the other piezo, which is consistently sampled. Additionally, both of these corresponding peaks show a moderate degree of negative correlation between the peak power and the average bellows position. Only the peaks at (69 ± 1) kHz and (67 ± 1) kHz have a relatively moderate number of sampled points, but exhibit a weak correlation with the average bellows position.

The consistent correlation of these frequency peaks with the position of the bellows suggests that they originate from the bellows. The stretching or unstretching of the bellows can cause changes in the distribution of power across the harmonic modes, and this, in turn, can alter the acoustic power response. Notably, the fact that the peaks at (69 ± 1) kHz in Piezo 0 and (67 ± 1) kHz in Piezo 19 are not coupled to the bellows in the same way further supports the idea that the observed effects are peak-dependent. This indicates that the other peaks we found are likely associated with the bellows.

4.2 Future Work

To improve our analysis and guide future research, we should seek a better understanding of our acoustic environment. One possible avenue is to test the frequency responses of the piezos, which may explain the discrepancies in our measured excitation frequencies. Clearly, from Table 1, we remain unsure as to why certain peaks are present in only one piezo and not the other. We also cannot explain the dip around 20 kHz seen in Fig. 2 in piezo 0. Without a clear understanding of how our piezos respond to sound, it's challenging to know how to attribute these effects. Furthermore, installing external piezos could also help us differentiate between lab noise and detector noise.

Additionally, we can also work to refine our Q factor analysis and decay constant estimation. Running COMSOL simulations to predict the expected Q factors would provide a comparison to our findings, as well as validate the frequencies we identified.

Finally, we should experiment with decreased segment length for our Welch method, or switch to a discrete Fourier transform (DFT) over the entire interval, to improve our spectral resolution. Currently, we are unable to resolve a beat frequency on the order of 100 Hz, which we believe we are seeing. By fitting sine functions to the oscillations observed in Fig. 4 and Fig. 5, we could isolate this beat frequency and analyze it in more detail. This would improve our understanding of the detector noise characteristics.

5 Conclusion

We set out to identify and characterize the resonant frequencies of the PICO-40L detector. By using the mechanical motion of the bellows during compression as an impulse to the bellows system, we successfully identified several frequency peaks between 25 kHz and 100 kHz. In Piezo 0, we observed peaks at (29 ± 1) kHz, (35 ± 1) kHz, (45 ± 1) kHz, (53 ± 1) kHz, and (88 ± 1) kHz. In Piezo 19, we identified peaks at (28 ± 1) kHz, (39 ± 1) kHz, (53 ± 1) kHz, (60 ± 1) kHz, (79 ± 1) kHz, (91 ± 1) kHz, and (100 ± 1) kHz, which are coupled to the bellows to a moderate to high degree. The discrepancies in peaks between the Piezos remains unexplained, but could possibly be due to piezo frequency response or piezo placement.

Furthermore, many of these peaks exhibit a beat frequency, suggesting that our detector is self-interacting and that several components are beating with each other. Further work can be done to analyze the beat frequencies, investigate the piezo frequency response, account for potential lab noise, and run finite element analysis to validate our findings.

6 Acknowledgements

The author thanks Dr. Carsten Krauss for his valuable insight and guidance during this work, as well as Victoria Burman and Jack MacArthur for their input and discussion.

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